

From Position Bias to Position-Stable Agreement: Testing What LLM Judges Evaluate

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Abstract

Pairwise LLM-as-judge evaluation should preserve response identity when presentation order changes. Across four judge models, short A/B prompts yield **70–100% second-position selection** (72/80 comparisons overall). Targeted prompt ablations show **format sensitivity**: label changes alter parsed choices, and a neutral buffer produces complete first-position selection on 59 fixed-order comparisons, reversing rather than removing position preference. We use a **position-swapped evaluation protocol**, retaining a directional verdict only when both orders select the same response. In a new 16-condition GPT-4.1 factorial audit, analysis-first and a verdict prefix raise directional coverage by 5.6 and 6.5 percentage points, respectively. On 40 Arena-Hard queries with three generators, the 120 pairs comprise 62 directional results, no double ties, 14 swap failures, and 44 invalid pairs. Cross-Model Agreement (CMA) yields 25 unique query winners. A retrospective domain-routing check predicts 17 of these 25 labels using rules fitted outside each test fold. These labels and any oracle score remain judge-relative; human-preference validity and downstream training gains are not established.

1 Introduction

Automated pairwise comparisons supply labels for RLHF (Ouyang et al., 2022; Stiennon et al., 2020), routing (Ong et al., 2025; Shnitzer et al., 2023), and rankings (Zheng et al., 2023; Li et al., 2023b). Useful verdicts should remain stable to irrelevant changes in response order or label format. Such stability checks content dependence, without proving factual correctness or human-preference validity (Guo et al., 2024).

Minimalist prompts select the second response on 70–100% of comparisons across four judges ($N = 20$ each). A separate ten-comparison audit gives 70% disagreement with swap-consistent ref-

erences, not measured human-label error. This sensitivity matters when labels support preference optimization (Rafailov et al., 2023) or benchmark rankings (Li et al., 2024). Order-balanced, tie-aware protocols provide a practical check (Zheng et al., 2023; Ye et al., 2024; Wang et al., 2024a).

FairEval and swap-based protocols in MT-Bench and Chatbot Arena already address position bias (Wang et al., 2024a; Zheng et al., 2023). We build on these audits to examine prompt-format sensitivity and the response-level signal surviving swap filtering, an ongoing concern in judge meta-evaluations (Chen et al., 2024a; Thakur et al., 2025; Huang et al., 2025). Our contributions are:

1. We document **format-sensitive verdicts**: short A/B prompts select the second response on 90% of 80 comparisons across four judge families. New factorial audits separate four prompt factors on two API judges and distinguish swap failures from format failures.
2. CMA separates **coverage and abstention**: 62 of 120 pairs retain a direction and 25 of 40 queries have a unique winner. Explicit ties, swap failures, invalid outputs, and query-level abstentions remain distinct.
3. **Candidate-pool dependence**: a nested two-to-three-generator analysis distinguishes selection of the new candidate, changes involving unresolved queries, and preservation of old comparisons. A separate retrospective routing check uses query-disjoint fitting and evaluation.

2 Related Work

Grounded and Faithful Evaluation. Grounding is often studied as a property of generation: an answer should be supported by retrieved evidence, source documents, tools, or other external context. We study a complementary validity question for

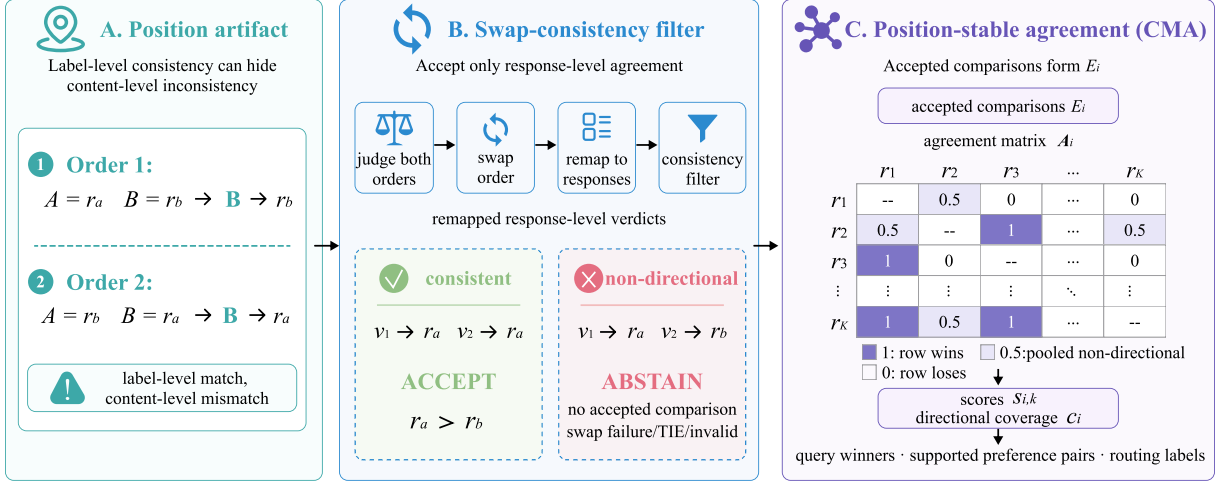


Figure 1: From position artifacts to position-stable agreement. The prompt requests analysis followed by a verdict. Position swapping remaps verdicts to response identities and retains a direction only when both orders select the same response. The illustrative matrix pools non-directional outcomes numerically; our reports retain explicit TIE, swap failure, and invalid categories separately. CMA summarizes the accepted comparisons for pool-specific evaluation and candidate supervision.

evaluation itself. In pairwise LLM-as-judge evaluation, a verdict should not change solely because the responses exchange positions or labels. We therefore use position invariance as a minimal necessary test of response dependence, while reserving “grounded” for stronger validation against response quality, evidence, or human judgments.

LLM-as-Judge and Position Bias. Automated pairwise evaluation is now the dominant quality signal for leaderboards (Zheng et al., 2023; Li et al., 2023b; Dubois et al., 2024), reward modeling (Ouyang et al., 2022; Rafailov et al., 2023), and routing (Ong et al., 2025). Prior work improves judge quality through fine-grained scoring (Ye et al., 2024), evaluator models (Zhu et al., 2025; Kim et al., 2024; Wang et al., 2024b), and debiasing (Wang et al., 2024a). FairEval explicitly counterbalances presentation order, while MT-Bench and Chatbot Arena use position swapping and permit ties (Wang et al., 2024a; Zheng et al., 2023). Our swap-and-filter rule builds on this literature: its distinct use is to convert response-identity consistency into an abstention-aware matrix for cross-model analysis. Meta-evaluations nevertheless show that minimalist prompts remain common (Chen et al., 2024a; Thakur et al., 2025; Huang et al., 2025).

Preference Learning and Multi-Model Systems. Preference-based training and self-improvement methods (Wu et al., 2025; Chen et al., 2024b; Yuan et al., 2024; Zhao et al., 2026; Shenfeld et al., 2026; Hübotter et al., 2026) inherit any bias in their evaluator. Multi-model routing and

ensemble systems instead exploit model diversity (Shnitzer et al., 2023; Jiang et al., 2023; Wang et al., 2025), while peer-evaluation methods use multi-agent consensus online (Li et al., 2023a; Chan et al., 2024). CMA differs by extracting offline routing and preference signals from swap-consistent response-level comparisons among diverse generators.

3 Method

3.1 Problem Setting

For prompts $\{q_i\}_{i=1}^N$ and models $\{M_k\}_{k=1}^K$, each model produces $r_{i,k} = M_k(q_i)$. We seek response-level preferences that are stable to A/B presentation labels. Position stability is a necessary validity check, not proof that the preference matches human judgment or factual quality. We aggregate accepted comparisons across $\{r_{i,1}, \dots, r_{i,K}\}$ for complementarity analysis, exploratory routing, and candidate preference-pair construction.

3.2 Position-Stabilized Pairwise Comparison

Our protocol turns a potentially position-sensitive A/B choice into a response-identity judgment.

Prompt Design. The comparison prompt requests an analysis before a structured verdict, uses 1/2 labels, and offers an explicit TIE option. The archived short-versus-reasoning comparison changes all four components together. New 2^4 factorial audits vary these components independently

on fixed responses with GPT-4.1 and DeepSeek-V3.2 (Appendices E–F); they test visible-analysis and format effects without attributing the archived difference to reasoning alone.

Position Swapping and Agreement Filtering. A reasoning-first prompt alone does not guarantee position stability. For each response pair (r_a, r_b) , we therefore:

1. Present r_a first, r_b second. Record verdict v_1 .
2. Present r_b first, r_a second. Record verdict v_2 .
3. Remap both verdicts back to response identities. Accept a directional label only if both parsed verdicts select the same response; otherwise abstain.

With numeric labels, verdicts $(1, 2)$ select r_a in both presentations, whereas $(2, 1)$ select r_b . Identical labels such as $(1, 1)$ select different responses and therefore fail the check. This distinction is essential: consistency is defined after remapping to response identity, not by matching the emitted tokens. A TIE in only one order also fails to establish a stable direction, even when the other verdict is valid.

Non-directional pair outcomes have three distinct causes: both parsed verdicts are TIE; both outputs parse but disagree after response-identity remapping, including a tie in only one order; or at least one output is missing or unparseable. We retain these cause labels separately even when the matrix represents all three by 0.5. Invalid takes precedence if either order lacks a usable verdict. Our 40-query $K = 3$ analysis preserves both parsed labels for every pair. Query-level abstention additionally includes tied aggregate scores, even with directional edges (Appendix C.1).

3.3 Cross-Model Agreement Structure

Given accepted pairwise comparisons among all K responses for q_i , we encode the response-level outcomes in an **agreement matrix** A_i :

$$A_i[j, k] = \begin{cases} 1 & \text{if } r_{i,j} \succ r_{i,k} \\ 0.5 & \text{otherwise} \\ 0 & \text{if } r_{i,j} \prec r_{i,k} \end{cases} \quad (1)$$

The average pairwise score for model M_k is

$$s_{i,k} = \frac{1}{K-1} \sum_{j \neq k} A_i[k, j] \quad (2)$$

The fraction of directional pair outcomes is

$$c_i = \frac{2}{\binom{K}{2}} \sum_{j < k} |A_i[j, k] - 0.5|. \quad (3)$$

Thus $c_i \in [0, 1]$ measures directional coverage, not agreement on a best response or transitivity: a three-response preference cycle has $c_i = 1$. We use the normalized directional-win count $w_{i,k} = (K-1)^{-1} \sum_{j \neq k} \mathbf{1}[A_i[k, j] = 1]$, and select a winner only if its count is uniquely largest. This differs from $s_{i,k}$, which gives half credit to every pooled pair outcome. Query-level abstention includes tied top counts and cannot be equated with a semantic tie.

Why the score definition matters. In an illustrative four-response graph, r_a beats r_b , r_b beats r_c and r_d , and the other three edges are pooled. Response r_b uniquely leads in directional wins, although it loses to r_a . Their half-credit scores are both $2/3$. Thus a unique winner under w need not be unique under s , nor preferred to every other response. This constructed example motivates checking individual edges before converting a query-level winner into pairwise supervision.

3.4 Preference Pairs and Routing

For a query q_i with a unique aggregate winner M_w , a target response from M_t forms a candidate preference pair:

$$(q_i, r_{i,w}, r_{i,t}) \rightarrow \text{preference pair} \quad (4)$$

Candidate preference pairs can be restricted by directional coverage $c_i > \tau_c$ and a positive score margin. Our sensitivity analysis uses $w_{i,k}$ and the top-versus-runner-up margin; its baseline requires a unique top directional-win count. For directional supervision we additionally require a saved accepted edge that directly favors the selected winner. A learned router could predict the selected model from query features $x(q)$:

$$\hat{k}(q) = \arg \max_k P(M_k \text{ wins} \mid x(q)) \quad (5)$$

Let U contain queries with a unique directional-win winner k_i^* . Winner-label accuracy is

$$\text{Acc}(h) = \frac{1}{|U|} \sum_{i \in U} \mathbf{1}[h(q_i) = k_i^*]. \quad (6)$$

Unresolved queries receive no label; report $|U|/N$ as coverage. A winner-label oracle scores one

on U . The mean half-credit score $R(h) = N^{-1} \sum_i s_{i,h(q_i)}$ instead includes pooled outcomes. Its oracle maximizes $s_{i,k}$ per query; a three-way cycle gives every response $s = 0.5$, so even this oracle scores 0.5. Uniform random selection has expected half-credit score 0.5 and winner-label accuracy $1/K$. Neither metric validates answer quality. We report winner-label accuracy for the new retrospective routing analysis and exclude historical routing percentages with unrecoverable scoring provenance.

4 Experiments

4.1 Experimental Setup

Models. Our primary setting uses **GPT-4o-mini**, **DeepSeek-V4-Flash**, and **Qwen3.5-35B-A3B**. These span three organizations. Model names and archived runner settings identify the tested configurations; the old records do not preserve independently verifiable weight snapshots.

Evaluation Queries and Judge. The primary Arena-Hard analysis (Li et al., 2024) uses all 40 queries in a recovered $K = 3$ batch with both-order parsed verdicts. The runner specifies GPT-4.1, temperature zero, full questions and generated responses, a 1,500-token generation budget, and a 250-token judge budget (Appendix A.3). The 120 response strings are fully stored, while raw judge text is limited to 500-character previews. Classification uses the separate parsed-verdict fields. A different 29-query subset supports the verbosity and factorial audits. These are available archival batches, not randomly sampled or interchangeable replications.

The earlier $K = 4$ archive lacks the two-order labels needed to distinguish reasons for abstention. We therefore exclude its aggregate winner, ranking, and threshold statistics from the revised empirical claims. We also exclude the older 100-query pool-expansion summary whose selection transitions cannot be recovered. The smaller analysis here replaces those results; it does not reconstruct missing historical judgments.

The experiments address three levels: individual verdict distributions, pairwise stability under position exchange, and query-specific winner selection. Their denominators are attempted verdicts, unordered pairs, and queries, respectively. We report these separately because more TIE verdicts can reduce directional-choice rates, while tied ag-

gregate scores can prevent winner selection even when some individual comparisons are directional.

4.2 Position Bias in LLM-as-Judge

Short forced-choice prompts select the second response on 18/20 comparisons for GPT-4.1, 14/20 for GPT-4o-mini, and 20/20 for each of DeepSeek-V4-Flash and Qwen3.5-35B-A3B (Appendix Table 10). The pooled rate is 72/80. A separate ten-comparison summary records 70% disagreement with swap-consistent references.

A one-order prompt audit retains enough information to distinguish ties from parser failures (Table 1). It evaluates the same ten queries with each judge, placing GPT-4o-mini first and DeepSeek-V4-Flash second, using stored response previews. Every directional verdict selects the second response. Consequently, a lower second-choice share among all parsed outputs reflects more TIE verdicts, without showing that directional position bias has been removed. Fixed model order also confounds position with generator identity and response content.

Judge	First	Second	TIE	Invalid
GPT-4.1	0	4	5	1
GPT-4o-mini	0	8	2	0
DS-V4-Flash	0	5	5	0
Qwen3.5-35B	0	6	2	2

Table 1: One-order prompt audit, ten inputs per judge. Counts are reconstructed from saved parsed verdicts. TIE is an explicit output in this audit; invalid denotes a missing parsed verdict. The audit contains no swapped presentation, so a swap-failure rate is unavailable.

Position swapping tests a different property: whether the same response wins in both presentations. A retained directional pair necessarily selects the first position once and the second position once. Its resulting 50/50 position count is a consequence of filtering, rather than an independent accuracy result. The useful quantities are directional coverage, the causes of abstention, and validity against an external quality reference.

4.3 Cross-Model Agreement Results

Query-Specific Winners. At $K = 3$, 25 of 40 queries have a unique top directional-win count (Figure 2). The 15 unresolved queries differ from non-directional pair outcomes: a query aggregates three comparisons, any combination of which may lack a direction (Table 2).

Outcome	Count	Share
Directional pairs	62/120	51.7%
Explicit TIE in both orders	0/120	0.0%
Swap-check failures	14/120	11.7%
Invalid pairs	44/120	36.7%
Unique-winner queries	25/40	62.5%
Unresolved queries	15/40	37.5%

Table 2: Separate pair-level causes and query-level outcomes in the 40-query $K = 3$ batch. Invalid means at least one missing or unparsed verdict. Unresolved means a tied top directional-win count, not explicit agreement on equal answer quality.

Routing Scope. Query-specific winners suggest hypotheses for routing (Ong et al., 2025; Lu et al., 2024). Section 5.7 evaluates a newly specified keyword-domain router with query-disjoint training and test folds. This retrospective check predicts the stored winner labels; it is distinct from evaluating human-rated answer quality, training a student, or recovering the historical router.

Candidate-Pool Comparison. Section 5.6 compares the first two generators with the complete three-generator pool on these same 40 queries. It reuses every saved edge and reports all selection transitions. The historical six-query $K = 5$ pilot appears only in the appendix.

Preference Pair Yield. The 25 unique-winner queries offer $25 \times 2 = 50$ potential winner-candidate pairs. Only 36 directly prefer the selected winner; three fail the swap check and eleven are invalid. We exclude those fourteen from directional supervision. A unique aggregate winner therefore does not imply an accepted preference against every candidate, and candidate supervision is not a validated training gain.

The gap between 50 candidate edges and 36 supported edges makes the aggregation issue operational. Assigning every comparison involving a selected winner a positive label would convert fourteen abstentions into preferences. Keeping the original edge categories avoids this change. Conversely, directional comparisons can remain informative on a query with no unique aggregate winner. The 62 accepted pairs and the 25 winner queries therefore describe different resources: local pairwise evidence and a more restrictive model-selection target.

5 Analysis

We next examine prompt sensitivity, agreement across judges, and dependence on the candidate pool. Supplementary one-order audits remain distinct from the main swap-filter protocol, and the routing check evaluates only its derived winner labels.

5.1 Cross-Judge Agreement

Qwen2.5-72B-Instruct agrees with the reference winner on 33/52 directional comparisons (63.5%). Of the 59 attempted comparisons, five return TIE and two are unparsed. This audit randomizes one presentation order and judges stored response previews; it does not repeat the main two-order protocol. A separate rubric-scoring summary reports agreement on 17/25 directional cases (68.0%), with 25 additional ties among 50 attempted cases. These checks provide limited cross-judge evidence, without validating the preferences against human judgments or factual correctness.

The rubric’s higher agreement also conditions on lower coverage: 25/50 attempts versus 52/59. Neither rate ranks validators without accounting for these retained subsets. Appendix Table 9 gives descriptive Wilson intervals, without establishing human-level accuracy.

5.2 Position Bias: Deeper Analysis

Short-Prompt Choices Across Judges. The four-judge summary gives 72/80 second-position selections (90%). Its pooled Wilson interval is [81.5%, 94.8%], treating the reported counts as binomial observations. Dependence between reused queries and judges is not represented by that interval. Appendix Table 10 gives the separate judge counts.

Prompt-Form Sensitivity. The Qwen2.5-72B ablations use 59 selected comparisons with the previously chosen winner always first, a 300-character query prefix, and up to 500 characters per response. The buffer changes second-label selection from 13/28 parsed outputs to 0/59, yielding complete first-position selection. Because order is fixed, this could reflect following the reference winner or a first-position shortcut; it does not establish elimination of bias. Numeric and multi-token conditions also change the parsed distribution, but their low parse coverage restricts interpretation. The multi-token condition additionally changes the output

budget from 5 to 100 tokens. These are prompt-form interventions with multiple possible explanations, not a factorial isolation of a token-association mechanism.

Condition	Parsed	Second label	Invalid
Short A/B	28/59	13/28 (46.4%)	31/59
Buffer A/B	59/59	0/59 (0.0%)	0/59
Numeric 1/2	19/59	14/19 (73.7%)	40/59
Multi-token A/B	12/59	12/12 (100%)	47/59

Table 3: Prompt-form audit, 59 fixed-order inputs per condition. Second-label rates condition on each row’s parsed subset. Invalid includes outputs rejected by the archived parser; raw completions are unavailable for reparsing. The conditions do not form a factorial design.

Parse coverage is part of the result. The multi-token condition’s 100% second-label rate describes only 12 parsed outputs, with 47 attempts invalid. The numeric rate uses 19 parsed outputs, whereas the buffer condition covers all 59. These changing subsets prevent a direct comparison on a common successfully judged sample.

Scope of the Archived Comparison. A further 59-input comparison yields 28 first, 12 second, and 19 invalid outputs for the short prompt; the reasoning-first prompt yields 20 first, 28 second, 10 ties, and 1 invalid. Second-choice share among directional outputs increases from 30.0% to 58.3%. This bundled comparison does not establish that analysis-first instructions alone reduce position preference.

Factorial Follow-up. We cross analysis-first versus verdict-only instructions, A/B versus 1/2 labels, TIE availability, and a VERDICT prefix on 87 full-response pairs from 29 queries. GPT-4.1 judges every condition in both orders (2,784 tasks), at temperature zero with a 500-token limit. Table 4 reports effects averaged over the other factors. Appendices E–F give every condition and interaction, including the separate DeepSeek-V3.2 audit on identical inputs.

On GPT-4.1, analysis-first raises directional coverage by 5.6 percentage points (95% interval [0.1, 11.5]) and reduces swap failures by 6.5 points. A VERDICT prefix adds 6.5 points of coverage ([2.6, 10.5]); its interaction with analysis-first is +6.0 points ([1.1, 10.6]). Invalid-pair rates change by +1.0 and −1.9 points for these factors, respectively. In the DeepSeek audit, analysis-first coincides with more format failures and nearly unchanged cover-

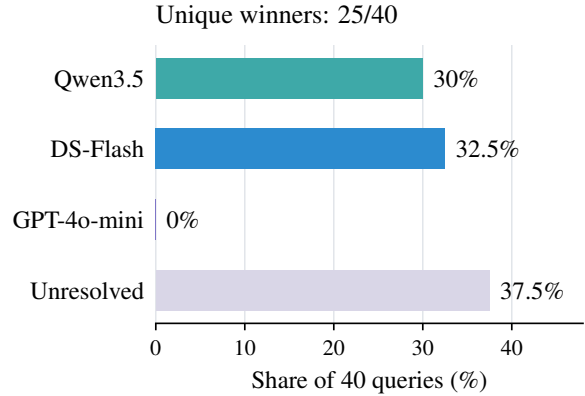


Figure 2: Winner distribution at $K = 3$ ($N = 40$), with the full query denominator. The unresolved category denotes tied top directional-win counts; it is distinct from an explicit pairwise TIE.

age (−0.7 points). These outcomes separate usable directions from swap and format failures.

Factor enabled	Directional	Swap failure
	+5.6	−6.5
Analysis first	[+0.1, +11.5] −2.4	[−12.5, −1.0] +0.4
1/2 labels	[−6.0, +1.3] −0.7	[−3.4, +4.3] −0.4
TIE allowed	[−2.7, +1.4] +6.5	[−2.7, +2.0] −4.5
VERDICT prefix	[+2.6, +10.5]	[−8.2, −0.7]

Table 4: Factorial main effects on GPT-4.1, in percentage points, averaged over the other three factors. Brackets: descriptive 95% query-cluster bootstrap intervals (29 queries, 2,000 draws; no multiplicity correction). All 87 pairs per condition remain in the denominator.

5.3 Generator Outcomes and Abstention

The primary batch evaluates three generators on 40 queries, yielding 120 unordered pair outcomes. Figure 2 shows 13 unique winners from DeepSeek-V4-Flash and 12 from Qwen; GPT-4o-mini supplies none. The 15 unresolved queries remain in the query denominator. This describes this tested pool, without establishing a scaling trend or general model ranking.

Abstention depends on the model pair. The DS-Flash–Qwen comparisons yield only 15 directional pairs, compared with 25 for GPT–DS-Flash and 22 for GPT–Qwen (Table 5). Their eleven swap failures account for most of the batch’s fourteen. Invalid pairs occur across all three pair types, so low directional coverage cannot be explained only by two similarly preferred generators.

Zero double ties does not establish that all re-

Pair	Dir.	TIE	Swap	Invalid
GPT-DS-Flash	25	0	1	14
GPT-Qwen	22	0	2	16
DS-Flash-Qwen	15	0	11	14

Table 5: Outcome causes by generator pair, with 40 attempted comparisons in each row. Dir. denotes a response-identity-consistent direction; Swap includes a TIE in only one order.

sponse qualities differ. It means that no pair receives an explicit TIE in both presentation orders. Among 240 order-level labels there is one TIE, paired with a directional choice in the other order; it is counted as a swap failure. The 51 missing or unparsed labels affect 44 pairs. The stored text previews do not identify whether those failures came from truncation, transport, or refusal.

5.4 Coverage Under Stricter Selection

The threshold sweep in Table 6 asks how much of the main batch remains when selection requires more directional evidence or a larger winning margin. Thresholds are applied strictly: $c_i > \tau_c$ and $w_{i,(1)} - w_{i,(2)} > \tau_m$. At $K = 3$, coverage changes in units of $1/3$ and the normalized win margin in units of $1/2$.

τ_c	τ_m	Clear	Edges	DS / Qwen
0	0	25/40	36/50	13 / 12
1/3	0	11/40	22/22	4 / 7
2/3	0	6/40	12/12	2 / 4
0	1/2	5/40	10/10	2 / 3
1/3	1/2	5/40	10/10	2 / 3
2/3	1/2	0/40	0/0	0 / 0

Table 6: $K = 3$ threshold sensitivity on 40 queries. Edges gives supported versus potential winner-candidate pair counts; 0/0 denotes an empty selected set, not a defined rate. GPT has no unique winners.

Requiring more than one directional edge reduces the selected set from 25 to 11 queries; all 22 winner-candidate edges in that subset are directly supported. Requiring all three comparisons to be directional leaves six unique-winner queries. A normalized winning margin above 0.5 instead selects five queries. These tests quantify coverage and separation in the judge’s graph; they have not been calibrated against external answer quality.

No query satisfies the strictest joint condition. This is also a structural constraint: with three fully directional edges, a transitive tournament has wins $(2, 1, 0)$ and margin $1/2$, while a cycle has tied

wins $(1, 1, 1)$. Neither exceeds $1/2$. Reporting the zero cell prevents interpreting a threshold sweep as a monotonic route to both higher confidence and usable labels. A query can have a larger margin only when some opponent comparison is non-directional.

5.5 Response-Length Association

Position-stable preferences can still be associated with verbosity. In the separate 29-query full-text subset, 52 of 59 directional comparisons select the longer response (Table 7). A judge can keep selecting that response after positions exchange, so this audit checks a different property from order consistency.

Quantity	Count	Rate	95% CI
Clear winner is longest response	15/20	75.0%	[53.1, 88.8]%
Pairwise preference selects longer response	52/59	88.1%	[77.5, 94.1]%
DS-Flash vs. Qwen selects longer response	14/19	73.7%	[51.2, 88.2]%

Table 7: Response-length association in the full-text $K = 3$ subset, with Wilson 95% confidence intervals. Counts condition on clear winners or directional comparisons. They describe association without isolating length from model identity or response quality.

GPT-4o-mini responses are shorter on average and rarely preferred, confounding length with model identity. The DS-Flash-Qwen subset still gives 14 longer-response choices among 19 directional pairs, but does not hold quality or content fixed. Length-controlled judgments would be needed to isolate verbosity. These counts use complete response strings from the 29-query subset, not the primary 40-query batch.

5.6 Same-Query Pool Extension

We start with GPT-4o-mini and DeepSeek-V4-Flash, then add Qwen on the same 40 queries using the stored two-order outcomes. The base generator pair is fixed by the model ordering, not selected for its transition rate. All old edges remain unchanged. The selected outcome changes on 24 queries, but Table 8 separates three explanations: a new-model winner, loss of an old unique winner, and resolution of a previously unresolved query in favor of an old model.

Only four changes directly replace an old unique winner with the new model. Eight select the new model from an unresolved starting point, while

Transition after adding Qwen	Queries
Old winner → new model	4
Unresolved → new model	8
Old winner → unresolved	10
Unresolved → old model	2
Old winner → different old model	0
Unchanged old winner	11
Unchanged unresolved outcome	5
Total	40

Table 8: Complete transitions from the fixed two-generator subpool to the three-generator pool. A new winner is defined by directional-win count, not independently validated answer quality.

twelve involve entering an unresolved outcome or resolving one in favor of an old model. Combining these transitions into a single “winner reversal” percentage would obscure the difference between model selection and pairwise stability.

Pool expansion can replace the selected winner or create or resolve a top-score tie while preserving every old comparison. A test of pairwise ranking instability would instead re-evaluate an old pair and observe a reversed direction. The absence of old-to-old replacements here is not evidence that such instability cannot occur: no old comparison is rejudged. The supplementary historical 20-query pilot is documented separately in Appendix D.1.

5.7 Retrospective Held-Out Routing Check

We evaluate a newly specified keyword-domain router in five folds of eight queries, shuffled with seed 20260913. Each fold uses only the other 32 queries’ resolved labels to fit domain modes and a global-majority constant baseline. Domains are code, math, science, and other; empty training domains fall back to the global mode. The fixed keywords, tie rules, split, and every fold result appear in Appendix D.2.

Every query receives one out-of-fold prediction. Accuracy is computed on the 25 unique-winner labels, with coverage $25/40 = 62.5\%$; the other 15 predictions are retained but not scored as correct or incorrect. The domain router matches 17/25 labels (68%), compared with 6/25 (24%) for the training-selected constant baseline. Fourteen labeled queries favor only the router and three favor only the baseline. Uniform random choice has expected label accuracy $1/3$, without requiring a sampled random run.

The labels are nearly balanced between DS-Flash (13) and Qwen (12), making the fitted ma-

jority baseline split-sensitive. Always selecting those models matches 13/25 and 12/25 labels; these are descriptive references, not test-selected baselines. The 17/25 result is a small retrospective label-prediction check, not a deployment or human-quality gain. The archive had already been inspected: folds separate fitting from evaluation but are not an untouched external test set.

6 Discussion

Evaluating every unordered pair in both orders requires $K(K - 1)$ judge calls per query. Six calls at $K = 3$, across sixteen prompt conditions and 29 queries, give 2,784 factorial tasks per judge. Pair subsampling or cheaper judges could reduce cost but would change the filter’s evidence. The two factorial batches record 5,568 successful tasks and nine API-failed attempts. Their reported token totals omit failed requests without usage; the archival batch also lacks complete billing records.

The protocol complements length-controlled evaluation (Dubois et al., 2024; Park et al., 2024): length control tests verbosity sensitivity, while swapping tests response-identity stability under order exchange. Both are needed when interpreting judge-derived preferences.

Practical guidance for position-stable evaluation.

Reports should specify the judge, prompt, verdict format, candidate pool, and swap rule; preserve both-order outputs; and separate explicit ties, swap failures, and invalid results. Query winners remain distinct from directly supported preference edges. Coverage determines label availability; external validation determines suitability for downstream use. Abstentions can trigger re-prompting or human review.

7 Conclusion

Short prompts show strong second-position selection in the archived audits, while factorial follow-ups separate four prompt components on GPT-4.1 and DeepSeek-V3.2. The 40-query $K = 3$ analysis distinguishes every pair’s recorded outcome cause and yields 25 unique winners. Pool extension changes selection without rejudging old edges; stricter thresholds reduce coverage. Position stability and retrospective winner-label prediction remain distinct from independently validated answer quality or training gains.

Limitations

Our experiments cover a limited set of models and datasets, which may constrain the generality of our findings. Future work should evaluate position stability across broader models, domains, and human-aligned criteria.

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Ethical Considerations

This work uses publicly available language models and benchmarks. No human subjects were involved. Candidate preference pairs reflect the tested judge’s preferences, which need not match human values, and should complement rather than replace human-in-the-loop alignment.

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A Evaluation Prompt Templates

A.1 Short Forced-Choice Prompt

Given the query below and two candidate responses, which response is better? Output ONLY "A" or "B".

Query: {query}

Response A: {response_a}

Response B: {response_b}

A.2 Neutral-Buffer Ablation Prompt

Which response is better? Output ONLY "A" or "B".

Query: {query}

Response A: {response_a}

Response B: {response_b}

Please take a moment to consider both responses carefully before making your choice. Remember that quality, accuracy, and helpfulness are the key criteria. Now, which is better?

A.3 Primary Three-Generator Judge Prompt

The 40-query runner uses this compact numeric prompt:

You are an impartial judge. Evaluate two responses to the question below.

[Question] {query}

[Response 1] {response_1}

[Response 2] {response_2}

Analyze strengths/weaknesses of each (2 sentences each), then state: VERDICT: 1 or VERDICT: 2 or VERDICT: TIE

The runner specifies GPT-4.1, temperature zero, a 1,500-token generation budget, and a 250-token judge budget. It passes the full question and the generated response strings, evaluates each pair in both orders, and stores each parsed VERDICT: label separately. It saves complete candidate strings but only 500-character judge previews. Generation finish reasons and resolved model snapshots were not retained; complete saved strings do not establish that generation never reached its token limit. These settings document the archived script, not a recovered full API trace.

A.4 Self-Evaluation Prompt

Rate the following response to the given query on a scale of 1-10, where 1 is terrible and 10 is perfect. Consider accuracy, completeness, clarity, and helpfulness.

Query: {query}

Response: {response}

Provide your rating as a single number.

A.5 Additional Prompt-Audit Details

The 59-input prompt-form audit uses the baseline:

Which response is better? Output ONLY "A" or "B".

Query: {query}

Response A: {response_a}

Response B: {response_b}

The numeric condition substitutes 1 and 2 for A and B throughout. The multi-token condition replaces the opening instruction with:

Which response is better and why? Start your answer with "Response A is better because" or "Response B is better because".

The buffer condition appends the exact text given above. All four run at temperature zero; the first three permit five output tokens and the multi-token condition permits 100. The script sends plain prompt strings to the local model. The archived parser examines the first five characters for A/B or 1/2, and the first 30 characters for "RESPONSE A" or "RESPONSE B" in the multi-token condition. Raw completions are unavailable, so parser rejection cannot be separated from output truncation.

The ten-query, four-judge audit instead uses:

You are an impartial judge. Evaluate two responses.

[Question] {query}

[Response 1] {response_1}

[Response 2] {response_2}

Analyze strengths/weaknesses of each (2 sentences each), then state: VERDICT: 1 or VERDICT: 2 or VERDICT: TIE

Stored response previews are supplied in one fixed model order. These prompt variants should not be treated as interchangeable replications.

B Additional Analysis

B.1 Five-Generator Pilot

A six-query historical pilot adds Gemini-3-Flash to a four-generator pool while reusing old pair outcomes. We retain only the separately logged new-model comparisons here; the pilot is too small to establish a scaling trend, and the older pooled edges prevent a full cause-level report for the five-generator pool.

The pilot saves both parsed verdicts for its 24 new-model pairs. Applying the outcome rule in Appendix C.1 gives 19 directional pairs, zero double-TIE pairs, four swap failures, and one invalid pair.

The four swap failures each contain a TIE in one order and a directional verdict in the other; the invalid pair has a missing verdict in one order. These counts concern only the pilot’s new-model edges and are not merged with the primary 40-query analysis.

B.2 Uncertainty and Threshold Sensitivity

Quantity	Count	Rate	95% CI
Short-prompt	72/80	90.0%	[81.5, 94.8]%
second-position preference			
Same-query winner	18/20	90.0%	[69.9, 97.2]%
change, pilot			
Cross-judge agreement, directional only	33/52	63.5%	[49.9, 75.2]%
Rubric-based agreement	17/25	68.0%	[48.4, 82.8]%

Table 9: Wilson 95% intervals from reported counts. They quantify binomial sampling uncertainty, without correcting for reused queries or judge dependence.

Table 9 reports Wilson intervals for the main count-based claims. The intervals are not hypothesis tests; they indicate the remaining uncertainty in finite query samples. The strongest conclusions are therefore qualitative and protocol-level: short prompts show high second-position preference, CMA abstains rather than manufacturing labels, and candidate-pool expansion changes, in practice, which systems win across otherwise fixed settings.

Section 5.4 presents the threshold sensitivity of the primary $K = 3$ batch and distinguishes potential pairs from supported directional edges. The new routing comparison is reported with exact counts and all folds rather than an independence-based significance claim.

B.3 Self-Evaluation Degeneracy

GPT-4o-mini rates its own outputs with near-uniform scores clustering at 8–9 out of 10 (mean 8.6, std 0.42, range 0.8 on the 10-point scale, $N = 10$ queries). In contrast, pairwise comparison within the same model does discriminate, producing clear $[0, 1, 2]$ point rankings across $K = 3$ samples. This pilot documents score compression in the tested absolute-scoring setup; it does not isolate the cause or establish a general failure of self-assessment.

B.4 Inference-Time Self-Improvement

We tested inference-time contrastive synthesis on four Arena-Hard queries: sample three responses

from one model, rank them, and synthesize a response. The synthesized output loses to greedy decoding on three queries and ties on one under the GPT-4.1 evaluation. This small negative pilot does not establish that greedy decoding is near-optimal or that external-model input is necessary for improvement.

C Supplementary Tables

Judge	Second choices	Rate	N
GPT-4.1	18/20	90%	20
GPT-4o-mini	14/20	70%	20
DeepSeek-V4-Flash	20/20	100%	20
Qwen3.5-35B-A3B	20/20	100%	20
Combined	72/80	90%	80

Table 10: Second-position selection under short forced-choice prompts. The rates exceed 50% for all four judges; these counts alone do not separate position from content preference.

C.1 Outcome Logging and Availability

Two-order evaluation should retain the raw completion and parsed verdict for both presentations, along with the query, response identities, prompt, model version, and generation settings. A deterministic cause assignment is: invalid if either order is missing or unparseable; explicit tie if both parse as TIE; directional if both select the same response after remapping; otherwise swap failure. This includes a tie in only one order as a swap failure.

The historical $K = 4$ archive does not preserve both-order verdicts. Its cause-specific counts are unavailable, not zero; we exclude its aggregate statistics from the revised empirical results. The one-order ten-query audit retains separate TIE and parser-failure counts (Table 1), but no swap-failure rate can be inferred without a second presentation.

Primary batch with both-order labels. The 40-query $K = 3$ batch preserves all 120 candidate-response strings and both parsed verdicts for each of its 120 unordered pairs. Its queries are disjoint from the excluded $K = 4$ batch. The saved labels recover 62 directional pairs, no double-TIE pairs, 14 swap failures, and 44 invalid pairs (Table 2). Thus its 58 non-directional outcomes reflect failed position consistency or unavailable verdicts, rather than explicit agreement that response quality is equal.

Of 240 saved order-level verdicts, 102 select the first position, 86 the second, one is TIE, and

51 are missing or unparsed. Among the 14 swap failures, 13 select the same position after swapping and one combines TIE with a directional choice. Only the first 500 characters of judge text are saved; classification therefore uses the recorded verdict fields rather than reparsing these previews. Missing finish reasons and complete API records prevent separating output truncation, transport failure, and semantic refusal. The batch supplies a reporting breakdown under one prompt configuration, not a factorial test or a reconstruction of the excluded historical batch.

D Detailed Per-Query Results

Table 11 reports every primary query, including unresolved outcomes.

D.1 Same-Query K-Scaling Details

Twenty stored $K = 3$ queries were extended with DeepSeek-V4-Pro. The old distribution was Qwen 9, DS-Flash 7, and unresolved 4; the new distribution was DS-Pro 16, unresolved 3, and DS-Flash 1. The selected winner changed on 18/20 queries (90%; Wilson 95% CI [69.9%, 97.2%]). The changes comprise 13 old-winner-to-new-model, 3 unresolved-to-new-model, and 2 old-winner-to-unresolved transitions. Original-model pair outcomes were reused; only the new pairs were judged, in one order. This pilot documents candidate replacement, without testing reversal of old pair preferences.

D.2 Retrospective Routing Protocol and Fold Results

This is a newly specified audit, not a reconstruction of the earlier domain router. Sort the 40 primary query IDs, shuffle with Python `random.Random(20260913)`, and assign successive IDs round-robin to five folds. Each fold contains eight queries. Only the other 32 queries can contribute labels to the domain mapping or constant baseline. We save the fold IDs, predictions for all queries including unresolved ones, and fitted mappings. The rules and split were fixed before computing this comparison; prior inspection of the archive is disclosed in the main text.

Domains match case-insensitive keywords with non-alphanumeric boundaries. The first matching domain in the following order is used; questions with no match are other.

Code: python, javascript, typescript, java, c++, sql,

ID	GPT	DS-F	Qwen	D	T	S	I	Winner
1	0	1	2	3	0	0	0	Qwen
3	0	0	1	1	0	1	1	Qwen
7	0	2	1	3	0	0	0	DS-F
9	0	0	0	0	0	0	3	–
11	0	1	1	2	0	0	1	–
13	0	1	0	1	0	0	2	DS-F
17	0	0	2	2	0	0	1	Qwen
19	0	1	1	2	0	1	0	–
21	0	1	1	2	0	1	0	–
23	0	1	2	3	0	0	0	Qwen
27	0	0	0	0	0	1	2	–
29	0	0	1	1	0	1	1	Qwen
31	0	1	0	1	0	0	2	DS-F
33	0	1	0	1	0	1	1	DS-F
37	0	0	0	0	0	1	2	–
39	0	2	0	2	0	0	1	DS-F
41	0	0	0	0	0	0	3	–
43	0	0	0	0	0	0	3	–
47	0	1	2	3	0	0	0	Qwen
49	0	1	2	3	0	0	0	Qwen
51	0	0	2	2	0	0	1	Qwen
53	0	1	1	2	0	1	0	–
57	0	1	0	1	0	0	2	DS-F
59	0	1	1	2	0	1	0	–
61	0	1	0	1	0	0	2	DS-F
63	0	0	2	2	0	1	0	Qwen
67	0	1	1	2	0	1	0	–
69	0	1	1	2	0	0	1	–
71	0	1	1	2	0	1	0	–
73	0	0	1	1	0	0	2	Qwen
77	0	1	0	1	0	0	2	DS-F
79	0	1	0	1	0	0	2	DS-F
81	0	1	0	1	0	0	2	DS-F
83	0	1	1	2	0	1	0	–
87	0	0	1	1	0	0	2	Qwen
89	1	1	0	2	0	0	1	–
91	0	2	1	3	0	0	0	DS-F
93	0	1	0	1	0	0	2	DS-F
97	0	2	0	2	0	1	0	DS-F
99	0	0	1	1	0	0	2	Qwen

Table 11: All 40 primary queries in archived Arena index order. GPT, DS-F, and Qwen columns give directional-win counts; divide by two to obtain $w_{i,k}$. D/T/S/I count directional, double-TIE, swap-failure, and invalid pairs. Each row has three pairs; – denotes no unique winner.

code, coding, program, programming, function, algorithm, debug, html, css, regex, array, string.

Math: math, mathematical, equation, integral, derivative, probability, theorem, prove, proof, algebra, geometry, matrix, calculate.

Science: physics, chemical, chemistry, biology, biological, molecule, quantum, cell, protein, genetic, medical, disease, patient.

Within each training fold, the global modal unique-winner label defines the constant baseline.

Each domain selects its modal training label, falling back to the global baseline when it has no labeled training examples. A domain-mode tie prefers the global baseline if it is a tied candidate; remaining ties use GPT, DS-Flash, then Qwen. Unresolved labels never enter fitting or accuracy denominators. This is a coarse reproducible text rule; it is not an independently annotated domain taxonomy.

Fold	Train $ U $	Test $ U $	Router	Const.	Baseline
1	20	5	5	0	Qwen
2	21	4	2	2	DS-F
3	21	4	3	1	Qwen
4	18	7	4	3	DS-F
5	20	5	3	0	DS-F
Total	—	25	17	6	—

Table 12: Five-fold retrospective routing check. Each fold has 32 training and eight test queries; $|U|$ counts those with unique-winner labels. Router and Const. are numbers of correct test predictions. Baseline names are chosen using training labels only.

Across all queries, code has eight scored labels, math four, science one, and other twelve. Router correct counts are 6, 1, 1, and 9; constant-baseline counts are 1, 2, 1, and 2. These small and uneven groups do not justify domain-general performance claims. Fixed-model reference accuracies are GPT 0/25, DS-Flash 13/25, and Qwen 12/25. We report all three as a diagnostic of the fitted baseline’s split sensitivity, without changing the frozen predictions or selecting a test-best model as the comparator.

D.3 Verbosity Audit

Position invariance does not rule out length bias. We therefore audit whether CMA preferences in the full-text $K = 3$ subset are associated with response length. This analysis uses 29 local result files that contain complete response text and pairwise verdicts; $K = 4$ files are excluded because their stored responses are truncated to 500-character previews.

Table 7 in the main text reports the query-level and pairwise associations. Its three denominators refer to the 20 queries with clear winners, 59 directional comparisons, and 19 directional DeepSeek–Qwen comparisons, respectively.

This audit should be interpreted as a limitation, not as a new quality metric. The association is partly confounded with model identity: GPT-4o-mini responses are much shorter on average and are rarely preferred in this subset, while DeepSeek-V4-Flash and Qwen3.5-35B-A3B have similar mean

lengths and split wins more evenly. The result shows that swap consistency addresses one presentation artifact, not all judge shortcuts. Future deployments should pair position-swap filtering with response-length controls whenever verbosity may shape verdicts.

E Factorial Prompt-Factor Audit

E.1 Fixed Inputs and Independent Factors

The supplementary audit uses the 29-query batch whose archive stores complete responses and merged pairwise_judgments. It includes all three pairs among GPT-4o-mini, DeepSeek-V4-Flash, and Qwen3.5-35B-A3B per query, without selecting pairs by archived verdict or swap outcome. This is an available-data subset, not a random sample of Arena-Hard. The separate 40-query batch in Appendix C.1 is not added after inspecting results. No candidate response is regenerated.

Four binary factors give 16 conditions: R requests analysis before the verdict rather than a verdict only; L uses 1/2 instead of A/B labels throughout; T allows an explicit TIE rather than forcing a direction; V prefixes the final label with VERDICT:. Every pair is evaluated in both orders in every condition. Queries, response strings, quality criteria, temperature zero, and the 500-token output budget are held fixed. The 2,784 requests are shuffled with seed 20260912 and sent with at most eight concurrent calls. R changes the request for visible analysis, not the position of a fixed analysis relative to a verdict.

The original design used GPT-4.1 as the default judge. Its archived routes were unavailable, so a first amendment fixed deepseek-v3.2 via Kingsoft Cloud before that batch began. The author subsequently supplied a working GreatRouter endpoint for gpt-4.1; a second amendment fixed that run before its formal results. The earlier DeepSeek results were already known. Every original input, condition, contrast, and parsing rule was retained, without selecting follow-up conditions by those results. Requested and returned model IDs are recorded and checked. The new GPT-4.1 batch estimates factor effects under its recorded settings, without independently verifying model weights or equivalence to the historical snapshot. DeepSeek results remain separate in Appendix F.

E.2 Exact Prompt Construction

Every request has one user message and no supplied system message. The common prefix is:

```
You are an impartial judge. Determine
which response better addresses the
question. Consider accuracy,
completeness, clarity, and helpfulness.

[Question]

{query}

[Response {first_label}]

{first_response}

[Response {second_label}]

{second_response}
```

The full query and saved responses replace these placeholders, separated by blank lines. Exchanging order exchanges the response strings, with their displayed labels still determined by position. Three instruction lines follow the second response. The first line is selected by R:

```
First analyze the strengths and
weaknesses of each response (two
sentences each), then give the final
verdict.
```

or

```
Give the final verdict only, without
analysis or explanation.
```

The second line is selected by T:

```
Use TIE when the two responses have
roughly equal quality.
```

or

```
Choose one response even when their
quality is close; TIE is not allowed.
```

The final instruction is:

```
The final nonempty line must be exactly
one of: {allowed_outputs}. Do not add
text after that line.
```

Allowed outputs use the two labels selected by L, followed by TIE when T is enabled. Each is prefixed with VERDICT: when V is enabled, and alternatives are joined by |. For example, condition 0111 uses VERDICT: 1 | VERDICT: 2 | VERDICT: TIE; condition 0000 uses A | B. The released prompt examples and raw requests preserve all 16 instantiated templates.

E.3 Parsing, Denominators, and Uncertainty

Parsing requires a normal stop finish, no provider-declared refusal, and an exact, case-sensitive final nonempty line. A verdict-only condition additionally requires a single nonempty output line. Format failures, empty outputs, and token-limited completions remain in the denominator. For each two-order pair, any invalid order yields an invalid pair; two TIE verdicts yield a double tie; opposite directional labels yield the same response identity and hence a directional pair; all other valid combinations are swap failures.

Pair rates divide by all 87 response pairs per condition. Raw first/second-position shares divide by all 174 order-level attempts, including ties and invalid outputs, so they are not conditional directional-choice probabilities. This makes changes in tie use and parse coverage visible alongside direction changes. The two-order filter is applied only after reporting the raw label counts.

Each main effect is the enabled-minus-disabled difference averaged over the eight settings of the other factors. Each two-factor interaction is a difference of differences averaged over four settings. We compute contrasts within each query and bootstrap the 29 query clusters with replacement for 2,000 draws using seed 20260912. The percentile 95% intervals are descriptive, with no adjustment for 60 reported contrasts. They measure query-sample variation, not repeated-run or provider variation.

E.4 GPT-4.1 Execution and Outcomes

All 2,784 formal tasks returned the expected gpt-4.1 model ID and stop finish reason. One HTTP 429 attempt and one read-timeout attempt were retained and resumed, yielding 2,786 API attempts. Concurrency fell from eight to four after the rate limit. No completion reported a length-limit finish or provider-declared refusal. Normally returned completions were never retried because of their verdict or formatting. The strict parser rejects 15 final-line format violations and two verdict-only outputs containing analysis, affecting 17 distinct pairs. Examples include 1 | 2, VER, and an otherwise valid final label preceded by unrequested analysis. A stop finish therefore does not establish compliance with the output contract.

Across all conditions, the 1,392 paired outcomes comprise 1,075 directional pairs, seven double ties, 293 swap failures, and 17 invalid pairs. The 2,784 raw outputs contain 1,453 first-position choices,

RLTV	Paired outcomes (87 per condition)				Raw order-level labels (174 per condition)			
	Directional	Double TIE	Swap failure	Invalid	First	Second	TIE	Invalid
0000	61	0	25	1	101	72	0	1
0001	70	0	17	0	98	76	0	0
0010	64	1	22	0	102	70	2	0
0011	66	1	20	0	100	72	2	0
0100	63	0	22	2	97	75	0	2
0101	65	0	22	0	103	71	0	0
0110	65	1	21	0	99	73	2	0
0111	64	1	20	2	100	69	3	2
1000	69	0	18	0	85	89	0	0
1001	75	0	12	0	91	83	0	0
1010	66	0	19	2	73	92	7	2
1011	75	0	12	0	88	84	2	0
1100	65	0	17	5	73	96	0	5
1101	72	0	15	0	88	86	0	0
1110	62	2	18	5	68	88	13	5
1111	73	1	13	0	87	81	6	0

Table 13: All 16 GPT-4.1 conditions on the same 87 response pairs. Bits encode analysis-first (R), numeric labels (L), TIE allowed (T), and a VERDICT prefix (V); 0 disables the factor and 1 enables it. Raw label counts precede swap filtering. A one-order TIE with a directional choice in the other order is a swap failure; any invalid order makes the pair invalid.

Contrast	Directional	Double TIE	Swap failure	Invalid pair
<i>R</i>	+5.6 [+0.1, +11.5]	−0.1 [−1.6, +0.9]	−6.5 [−12.5, −1.0]	+1.0 [−0.3, +2.2]
<i>L</i>	−2.4 [−6.0, +1.3]	+0.4 [0.0, +0.9]	+0.4 [−3.4, +4.3]	+1.6 [+0.6, +2.6]
<i>T</i>	−0.7 [−2.7, +1.4]	+1.0 [+0.1, +2.3]	−0.4 [−2.7, +2.0]	+0.1 [−0.9, +1.1]
<i>V</i>	+6.5 [+2.6, +10.5]	−0.1 [−0.7, +0.3]	−4.5 [−8.2, −0.7]	−1.9 [−3.6, −0.4]
<i>R</i> × <i>L</i>	−2.6 [−8.6, +3.7]	+0.9 [0.0, +1.7]	+0.3 [−5.7, +6.0]	+1.4 [−0.6, +3.4]
<i>R</i> × <i>T</i>	−1.4 [−5.7, +2.9]	−0.3 [−3.2, +1.7]	+0.9 [−3.7, +5.5]	+0.9 [−1.4, +3.4]
<i>R</i> × <i>V</i>	+6.0 [+1.1, +10.6]	−0.3 [−1.4, +0.6]	−2.6 [−6.9, +2.3]	−3.2 [−5.5, −0.9]
<i>L</i> × <i>T</i>	+0.9 [−3.7, +5.5]	+0.9 [0.0, +1.7]	−1.4 [−6.0, +2.9]	−0.3 [−2.3, +1.7]
<i>L</i> × <i>V</i>	−2.0 [−7.5, +3.4]	−0.3 [−1.4, +0.6]	+4.3 [−0.9, +9.8]	−2.0 [−4.3, +0.3]
<i>T</i> × <i>V</i>	−0.9 [−6.3, +4.6]	−0.3 [−1.4, +0.6]	+0.3 [−5.2, +5.5]	+0.9 [−1.1, +2.9]

Table 14: Factorial effects on paired-outcome rates, in percentage points, with descriptive 95% query-cluster bootstrap intervals. R, L, T, V follow Table 13. Main effects average enabled-minus-disabled differences; interactions average differences of differences over the remaining factors. Pair rates divide by all 87 pairs per condition, including invalid pairs. Intervals are not adjusted for the 60 reported contrasts.

Contrast	First position	Second position
<i>R</i>	−10.6 [−13.6, −7.8]	+8.7 [+5.9, +11.6]
<i>L</i>	−1.7 [−4.0, +0.4]	+0.1 [−2.0, +2.2]
<i>T</i>	−1.4 [−3.5, +0.6]	−1.4 [−3.3, +0.6]
<i>V</i>	+4.1 [+2.5, +5.9]	−2.4 [−4.0, −0.8]
<i>R</i> × <i>L</i>	−2.7 [−6.2, +0.4]	+0.7 [−2.3, +3.9]
<i>R</i> × <i>T</i>	−3.3 [−6.9, +0.4]	+0.1 [−3.2, +3.4]
<i>R</i> × <i>V</i>	+7.6 [+3.9, +11.4]	−4.2 [−7.8, −0.6]
<i>L</i> × <i>T</i>	+0.7 [−2.4, +4.0]	−2.2 [−4.9, +0.6]
<i>L</i> × <i>V</i>	+3.6 [+1.3, +6.0]	−2.4 [−4.5, −0.7]
<i>T</i> × <i>V</i>	+1.3 [−0.9, +3.3]	−0.1 [−2.7, +2.6]

Table 15: Factorial effects on unfiltered position-choice shares, in percentage points, with descriptive 95% query-cluster bootstrap intervals. R, L, T, V follow Table 13. Main effects average enabled-minus-disabled differences; interactions average differences of differences over the remaining factors. Position shares divide by all attempted orders, including ties and invalid outputs. Intervals are not adjusted for the 60 reported contrasts.

1,277 second-position choices, 37 TIE labels, and 17 invalid outputs (Table 13). These pooled counts summarize repeated conditions on the same pairs;

the independent sampling units for uncertainty remain the 29 queries.

Analysis-first yields 557/696 directional pairs versus 518/696 under verdict-only instructions. Invalid pairs are 12/696 and 5/696, respectively. Adding the prefix within analysis-first conditions raises directional coverage from 262/348 to 295/348; under verdict-only instructions it changes coverage from 253/348 to 265/348. Their difference of differences is the +6.0-point interaction. Numeric labels and TIE availability have directional-coverage main effects of -2.4 and -0.7 points, with descriptive intervals including zero. The full tables retain every contrast and outcome category, including formatting and raw position shares.

F Additional-Judge Factorial Results

The earlier DeepSeek-V3.2 batch uses the identical 29 queries, 87 response pairs, 16 conditions, prompt templates, and analysis in Appendix E. Its results are preserved separately from the GPT-4.1 follow-up. Model-specific output behavior can change the balance between directional coverage, ties, swap failures, and invalid results even under a shared output contract.

All 2,784 DeepSeek tasks returned a completion with the requested model ID and a normal stop, with no reported length-limit finishes or provider-declared refusals. Seven read-timeout attempts were retained in the log and retried, yielding 2,791 API attempts in total. Normally returned completions were never retried for their verdict or formatting. The strict parser rejects 87 outputs, all in analysis-first conditions, affecting 76 paired outcomes. Examples include `**1**` and `A | B`, rather than the required single final label. These are output-contract failures, not established failures to evaluate response quality; no parser change is made after observing them.

Analysis-first instructions reduce swap failures by 10.3 percentage points but increase invalid pairs by 10.9 points, leaving directional coverage nearly unchanged: -0.7 points (95% interval $[-5.3, 4.3]$). A VERDICT prefix reduces invalid pairs by 8.9 points and adds 5.9 points to directional coverage ($[1.3, 10.9]$). Its directional-coverage interaction with analysis-first is $+19.3$ points ($[10.1, 29.3]$). Fewer swap failures therefore need not produce more usable labels.

Across the eight analysis-first conditions, 76/696

pairs are invalid, compared with 0/696 under verdict-only instructions. Adding the prefix within analysis-first conditions raises directional coverage from 207/348 to 261/348; under verdict-only instructions it changes coverage from 243/348 to 230/348. Their 19.3-point interaction shows how bundled comparisons can conceal offsetting analysis and syntax effects. Numeric-label and TIE coverage intervals include zero (Table 17).

RLTV	Paired outcomes (87 per condition)				Raw order-level labels (174 per condition)			
	Directional	Double TIE	Swap failure	Invalid	First	Second	TIE	Invalid
0000	64	0	23	0	108	66	0	0
0001	59	0	28	0	115	59	0	0
0010	61	0	26	0	113	61	0	0
0011	57	1	29	0	115	57	2	0
0100	57	0	30	0	117	57	0	0
0101	57	0	30	0	117	57	0	0
0110	61	0	26	0	113	61	0	0
0111	57	1	29	0	115	57	2	0
1000	51	0	18	18	92	63	0	19
1001	64	0	19	4	105	64	0	5
1010	45	1	18	23	76	61	10	27
1011	63	1	22	1	99	64	10	1
1100	55	0	18	14	97	61	0	16
1101	67	0	19	1	104	69	0	1
1110	56	0	17	14	88	65	4	17
1111	67	1	18	1	97	72	4	1

Table 16: All 16 DeepSeek-V3.2 conditions on the same 87 response pairs. Bits encode analysis-first (R), numeric labels (L), TIE allowed (T), and a VERDICT prefix (V); 0 disables the factor and 1 enables it. Raw label counts precede swap filtering. A one-order TIE with a directional choice in the other order is a swap failure; any invalid order makes the pair invalid.

Contrast	Directional	Double TIE	Swap failure	Invalid pair
<i>R</i>	−0.7 [−5.3, +4.3]	+0.1 [0.0, +0.4]	−10.3 [−14.8, −5.7]	+10.9 [+7.6, +14.2]
<i>L</i>	+1.9 [−1.7, +5.3]	−0.1 [−0.4, 0.0]	+0.6 [−2.3, +3.7]	−2.3 [−4.0, −0.4]
<i>T</i>	−1.0 [−4.2, +1.9]	+0.7 [0.0, +2.0]	0.0 [−2.9, +3.0]	+0.3 [−1.4, +2.2]
<i>V</i>	+5.9 [+1.3, +10.9]	+0.4 [−0.4, +1.7]	+2.6 [−2.0, +7.5]	−8.9 [−12.5, −5.7]
<i>R</i> × <i>L</i>	+8.9 [+2.0, +15.8]	−0.3 [−0.9, 0.0]	−4.0 [−10.1, +2.3]	−4.6 [−8.0, −0.9]
<i>R</i> × <i>T</i>	−1.4 [−6.6, +3.7]	+0.3 [0.0, +0.9]	+0.6 [−3.4, +4.3]	+0.6 [−2.9, +4.3]
<i>R</i> × <i>V</i>	+19.3 [+10.1, +29.3]	−0.3 [−0.9, 0.0]	−1.1 [−7.2, +5.2]	−17.8 [−25.0, −11.5]
<i>L</i> × <i>T</i>	+4.9 [−0.9, +11.2]	−0.3 [−0.9, 0.0]	−4.0 [−9.5, +1.1]	−0.6 [−4.9, +3.4]
<i>L</i> × <i>V</i>	−0.9 [−7.8, +5.7]	+0.3 [0.0, +0.9]	−2.3 [−8.0, +3.7]	+2.9 [−1.4, +7.2]
<i>T</i> × <i>V</i>	+0.3 [−4.0, +5.2]	+0.9 [−0.9, +3.4]	+1.1 [−3.7, +5.7]	−2.3 [−5.5, +0.6]

Table 17: DeepSeek-V3.2 factorial effects on paired-outcome rates, in percentage points, with descriptive 95% query-cluster bootstrap intervals. *R*, *L*, *T*, *V* follow Table 16. Main effects average enabled-minus-disabled differences; interactions average differences of differences over the remaining factors. Pair rates divide by all 87 pairs per condition, including invalid pairs. Intervals are not adjusted for the 60 reported contrasts.

Contrast	First position	Second position
<i>R</i>	−11.1 [−14.2, −8.3]	+3.2 [+0.7, +5.9]
<i>L</i>	+1.8 [0.0, +3.7]	+0.3 [−1.5, +2.0]
<i>T</i>	−2.8 [−5.4, −0.6]	+0.1 [−1.4, +1.6]
<i>V</i>	+4.5 [+1.5, +7.8]	+0.3 [−2.4, +2.9]
<i>R</i> × <i>L</i>	+0.4 [−2.3, +3.0]	+3.7 [+1.3, +6.0]
<i>R</i> × <i>T</i>	−5.3 [−7.9, −2.9]	+1.1 [−1.6, +3.6]
<i>R</i> × <i>V</i>	+5.9 [+2.7, +8.9]	+4.9 [+1.0, +8.8]
<i>L</i> × <i>T</i>	−0.7 [−4.3, +2.4]	+2.9 [−0.3, +6.2]
<i>L</i> × <i>V</i>	−3.9 [−8.5, 0.0]	+2.6 [−0.9, +6.5]
<i>T</i> × <i>V</i>	+1.3 [−2.2, +4.5]	0.0 [−2.4, +2.7]

Table 18: DeepSeek-V3.2 factorial effects on unfiltered position-choice shares, in percentage points, with descriptive 95% query-cluster bootstrap intervals. *R*, *L*, *T*, *V* follow Table 16. Main effects average enabled-minus-disabled differences; interactions average differences of differences over the remaining factors. Position shares divide by all attempted orders, including ties and invalid outputs. Intervals are not adjusted for the 60 reported contrasts.